# Bohan Lu

*PhD Candidate, Quantum Error Correction*

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## Research Interests

Fault-tolerant quantum computing, especially transversal non-Clifford gates, quantum LDPC and Floquet codes, and decoding. Recent work uses projective geometry and SAT to construct codes implementing logical CCZ through transversal physical $T/T^\dagger$ gates.

## Education

2023–present **PhD in Electrical and Computer Engineering**, *Duke University*, Durham, NC, expected 2027–2028
Advisor: Kenneth R. Brown

2019–2022 **MSc in Physics**, *Freie Universität Berlin*, Berlin, Germany
Thesis: *Entanglement dynamics of a many-body localized system*. Advisor: Jens Eisert

2015–2019 **BA in Mathematics and Physics**, *Washington University in St. Louis*, St. Louis, MO
Cum Laude, Distinction in Mathematics, High Distinction in Physics

## Research Experience

2023–present **Doctoral Research, Brown Group**, *Duke University*, Durham, NC

- Constructed CSS codes implementing logical CCZ via physical transversal $T/T^\dagger$ using projective geometry and SAT, and proved $n \geq 39$ for CSS codes with 3 logical qubits, $d_Z \geq 3$, and quasi-transversal CCZ
- Designed a dynamical code from a minimally non-local hypergraph product code
- Developed stim-floquet (github.com/jerrylvx/stim-floquet), an open-source Python package for Floquet schedules, GF(2) homology checks, local detector discovery, and Stim circuit output

Oct 2025–May 2026, Aug 2026–present **Consulting Contractor**, *Xanadu Quantum Technologies*, Remote

- Research fault-tolerant quantum architecture (supervisor: Eric Sabo)

May–Aug 2026 **Architecture Intern**, *Xanadu Quantum Technologies*, Remote

- Researched decoding for quantum LDPC codes (supervisor: Eric Sabo)

Jul–Oct 2022 **Research Intern**, *HQS Quantum Simulations GmbH*, Karlsruhe, Germany

- Developed and benchmarked a Hermite-polynomial spectral method for resonant-level models and wrote an internal report (supervisor: Peter Schmitteckert)

2021–2022 **Master's Thesis Research**, *Freie Universität Berlin*, Berlin, Germany

- Co-developed an entanglement-based method to extract spatial properties of quasi-local integrals of motion, published in Communications Physics
- Co-wrote simulation code and ran tensor-network simulations on clusters

## Publications

2026 **Building codes with transversal CCZ using projective geometry and SAT solvers. B. Lu**, K. R. Brown. Preprint, arXiv:2610.09341. Code and data: github.com/jerrylvx/gate2code-public

2024 **Measuring out quasi-local integrals of motion from entanglement. B. Lu**, C. Bertoni, S. J. Thomson, J. Eisert. *Communications Physics* **7**, 17 (2024). doi:10.1038/s42005-023-01478-5, arXiv:2301.01787

## Skills

AI skills Agentic orchestration, engineering, and harness building

Computing Python, C++, HPC clusters

## Talks

Oct 2026 *Building codes with transversal CCZ using projective geometry and SAT solvers.* Applied Algebra Seminar, Virginia Tech (online), Oct 22, 2026 (upcoming).

Sep 2026 *Boolean functions: applications in quantum error correction and beyond.* Duke Quantum Center Theory Seminar, Sep 16, 2026.

2019 *Reduced density operators of finite-sized Heisenberg XXZ chains.* With A. Seidel. APS March Meeting 2019.

## Posters

2026 *Discovery and classification of small CSS codes supporting a quasi-transversal CCZ logical gate.* QEC 2026.

2025 *A dynamical error-correcting code from a minimally non-local hypergraph product code.* With K. Brown. QIP 2025 and QEC 2025.

2023 *Improved decoding of quantum LDPC codes using neural-network enhanced belief propagation.* With A. Pesah, J. Roffe, N. Raveendran. QIP 2023, QEC 2023 (Sydney), and *Coping with Errors in Scalable Quantum Computing Systems*, WE-Heraeus Seminar (Bad Honnef).

2022 *Entanglement dynamics in a many-body localized system.* With C. Bertoni, S. Thomson, J. Eisert. CRC 183 Summer School 2022.

## Teaching

Spring 2026 **Teaching Assistant, Liberata**, *Duke Bass Connections*, Duke University
- Mentored student research projects and contributed to research and development on the algorithms team

Fall 2024 Teaching Assistant, ECE 520 Quantum Engineering, Duke University

Winter 2020 Teaching Assistant, tensor-network seminar, Freie Universität Berlin

## Awards and Certificates

2026 Duke Graduate School Conference Travel Award

2025 NVIDIA Academy certificate, AI Infrastructure and Operations Fundamentals

2017–2018 Greg Delos Fellowship, Physics Department, Washington University in St. Louis

2016 Undergraduate Research Fellowship, Physics Department, Washington University in St. Louis

2015–2019 Dean's List, Washington University in St. Louis (Fall 2015 to Fall 2016, Spring 2019)